

PARAKOU, Benin

WMO: 653300

Lat: 9.350N

Lon: 2.617E

Elev: 1289

StdP: 14.02

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	66.9	68.5	22.2	17.5	77.8	25.9	20.8	78.8	14.1	74.3	12.2	75.8	4.4	40	0.301

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	20.3	98.8	71.5	97.2	71.3	95.8	71.1	78.9	89.9	78.2	88.9	77.5	87.9	4.5	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.0	142.4	84.3	75.3	139.3	83.5	74.7	136.3	82.8	43.4	90.5	42.7	89.4	42.1	88.0	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
13.5	11.7	10.2	DB	62.8	102.8	2.3	1.9	61.1	104.1	59.8	105.2	58.5	106.3	56.9	107.6
			WB	46.9	81.2	1.7	2.1	45.7	82.7	44.7	83.9	43.8	85.1	42.6	86.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	80.9	80.6	84.5	86.4	84.7	81.9	79.5	77.7	76.9	77.8	79.5	81.6	80.6	(6)
(7)		DBStd	3.71	3.27	2.70	2.24	2.83	2.57	2.33	2.05	1.90	1.90	1.94	1.80	2.34	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	11296	949	965	1129	1040	987	885	858	835	834	916	949	949	(10)
(11)		CDD65	5821	484	545	664	590	522	435	393	370	384	451	499	484	(11)
(12)		CDH74	62108	5710	7430	9267	7527	5896	4002	2652	2111	2674	3940	5458	5441	(12)
(13)		CDH80	27802	2804	4105	5258	3864	2511	1280	565	334	623	1337	2530	2591	(13)
(14)	Wind	WSAvg	4.5	4.6	4.6	5.1	5.5	5.1	4.6	4.3	4.3	3.5	3.7	3.9	4.9	(14)
(15)	Precipitation	PrecAvg	46.4	0.1	0.4	1.4	3.2	5.2	6.7	7.6	8.4	8.5	3.8	0.4	0.1	(15)
(16)		PrecMax	63.6	1.3	2.7	5.4	8.6	13.2	12.9	13.8	17.6	14.2	8.5	4.4	1.7	(16)
(17)		PrecMin	25.3	0.0	0.0	0.0	0.5	1.9	1.5	2.2	2.0	2.8	0.7	0.0	0.0	(17)
(18)		PrecStd	8.8	0.2	0.7	1.3	1.7	2.3	2.6	3.1	3.3	2.4	2.0	0.9	0.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	97.6	100.1	101.2	100.1	95.3	91.4	88.1	87.0	88.5	91.8	95.3	95.6	(19)
(20)			MCWB	65.4	69.6	71.4	73.9	76.9	76.9	76.2	76.3	76.7	74.5	69.5	65.8	(20)
(21)		2%	DB	94.8	97.8	99.1	97.7	92.8	89.3	86.1	84.9	86.6	89.8	93.6	93.7	(21)
(22)			MCWB	65.4	69.4	71.7	74.3	77.1	76.7	75.7	75.7	76.3	75.2	69.2	66.4	(22)
(23)		5%	DB	93.0	96.1	97.6	95.7	91.1	87.8	84.8	83.4	85.1	88.2	92.1	92.1	(23)
(24)			MCWB	66.0	68.9	71.9	75.3	77.0	76.3	75.4	75.3	75.9	75.6	69.7	66.6	(24)
(25)		10%	DB	91.2	94.6	96.2	93.5	89.6	86.2	83.3	81.8	83.7	86.5	90.5	90.6	(25)
(26)			MCWB	65.3	68.5	72.1	75.9	76.8	76.0	75.0	74.8	75.6	75.8	70.3	66.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.6	77.3	78.4	79.7	79.9	78.9	77.4	77.4	78.3	78.7	78.0	75.7	(27)
(28)			MCDWB	89.2	92.8	93.1	91.7	89.9	87.7	85.1	84.5	85.8	87.1	87.3	87.5	(28)
(29)		2%	WB	73.6	75.9	77.3	78.6	78.8	77.7	76.6	76.4	77.2	77.8	76.7	74.3	(29)
(30)			MCDWB	86.3	89.7	91.7	91.7	89.4	86.5	83.9	83.0	84.6	86.1	87.2	85.8	(30)
(31)		5%	WB	72.1	74.7	76.5	77.9	78.1	77.1	76.0	75.6	76.5	77.2	75.7	73.0	(31)
(32)			MCDWB	84.7	88.0	90.5	90.7	88.4	85.8	83.0	81.9	83.5	85.5	85.6	84.3	(32)
(33)		10%	WB	70.4	73.7	75.7	77.1	77.3	76.4	75.4	75.1	75.8	76.5	74.7	71.7	(33)
(34)			MCDWB	83.8	87.2	89.1	89.6	87.4	84.7	82.0	80.9	82.5	84.5	84.1	82.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	22.1	21.2	20.3	18.0	15.1	12.9	10.9	10.0	12.0	14.8	20.0	22.2	(35)
(36)			MCDBR	23.9	22.9	22.6	21.0	17.5	15.3	13.3	12.4	14.6	17.2	21.8	23.6	(36)
(37)			MCWBR	11.8	10.1	7.3	5.6	5.6	5.3	4.9	4.8	6.1	6.5	7.3	10.3	(37)
(38)		5% WB	MCDWB	20.1	20.5	19.7	18.0	15.6	13.9	12.2	11.3	13.5	15.2	17.7	19.9	(38)
(39)			MCWBR	11.5	8.7	6.1	6.0	6.1	5.5	5.0	4.9	6.2	6.7	6.2	9.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.709	0.757	0.803	0.706	0.607	0.554	0.508	0.495	0.507	0.588	0.593	0.613	(40)	
(41)		taud	1.603	1.538	1.475	1.680	1.855	1.996	2.167	2.206	2.152	1.893	1.821	1.774	(41)	
(42)		Ebn at Noon	204	200	194	212	229	238	251	258	258	236	230	223	(42)	
(43)		Edh at Noon	84	92	99	80	66	56	48	47	50	64	67	69	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1658	1722	1826	1843	1759	1594	1386	1282	1463	1650	1753	1654	(44)	
(45)		RadStd	48	50	72	50	70	63	59	80	73	77	51	63	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page